

**THEORY COURSE OUTLINE**

1.	Faculty	Faculty of Engineering & Applied Sciences (FEAS)															
2.	Department	Electrical and Electronic Engineering (EEE)															
3.	Program	B.Sc. in CSE															
4.	Course Title	Electrical Technology															
5.	Course Code	EEE 101															
6.	Trimester	Fall 2024															
7.	Pre-requisites	N/A															
8.	Course Type	Interdisciplinary Course															
9.	Credit Hours	3.0															
10.	Intake/Section	53/S1 (Day)															
11.	Class Schedule	<table><tr><th>Intake/Section</th><th>Class Day</th><th>Class Hours</th><th>Venue</th></tr><tr><td rowspan="2">53/S1 (Day)</td><td>Sunday</td><td>05.15 pm - 06.30 pm</td><td>B2/906</td></tr><tr><td>Tuesday</td><td>09.15 am - 10.30 am</td><td>B2/906</td></tr></table>				Intake/Section	Class Day	Class Hours	Venue	53/S1 (Day)	Sunday	05.15 pm - 06.30 pm	B2/906	Tuesday	09.15 am - 10.30 am	B2/906	
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12.	Campus	Permanent Campus															
13.	Course Website																
14.	Course Instructor Details	<table><tr><td colspan="2">Name: Md. Sabbir Hasan Sohag</td><td colspan="2">Designation: Assistant Professor, EEE, BUBT</td></tr><tr><td colspan="4">Specialization: Electronics and Photonics, Communication, HVDC & IoT</td></tr><tr><td colspan="2">Room No. B3/ 608</td><td>Email: s.hasan@bubt.edu.bd</td><td>Cell No. 01747697159</td></tr></table>				Name: Md. Sabbir Hasan Sohag		Designation: Assistant Professor, EEE, BUBT		Specialization: Electronics and Photonics, Communication, HVDC & IoT				Room No. B3/ 608		Email: s.hasan@bubt.edu.bd	Cell No. 01747697159
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15.	Counselling Schedule /Tutorial Class	<table><tr><th>Intake/Section</th><th>Day</th><th>Counseling Hours</th><th>Venue</th></tr><tr><td rowspan="2">52/S6 (Day)</td><td>Tuesday</td><td>02.00 pm - 03.15 pm</td><td>Room: B3/ 608</td></tr><tr><td>Thursday</td><td>02.00 pm - 03.15 pm</td><td>Room: B3/ 608</td></tr></table>				Intake/Section	Day	Counseling Hours	Venue	52/S6 (Day)	Tuesday	02.00 pm - 03.15 pm	Room: B3/ 608	Thursday	02.00 pm - 03.15 pm	Room: B3/ 608	
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16.	Text Book	1. Fundamentals of Electric Circuits: <i>C. Alexander, M.N.O. Sadiku, 7th Ed., McGraw Hill</i>															
17.	Reference Book	1. Electric Circuits: <i>J.W. Nilsson, S. Riedel, 11th Edition, Pearson.</i> 2. Introduction to Electric Circuits: <i>J.A. Svoboda, R.C. Dorf, Wiley & Sons</i> 3. Introductory Circuit Analysis: <i>R. L. Boylestad, 13th Edition, Pearson</i>															
18.	Equipment & Aids	Students must carry learning materials (<i>calculator, notebook, pen, etc.</i>) in classroom. Borrowing learning materials in exam or classroom from fellow students are strictly prohibited. <i>Students are also advised to keep at least one sheet of blank paper (A4 size) with them every class.</i>															
19.	Course Rationale	This is a core course of Computer Science and Engineering (CSE) program that aims to introduce the students with fundamental methods and theorems of circuit analysis. Students will learn about basic electrical parameters and their units, electrical devices and their applications and use first laws of circuit analysis to solve for those quantities and analyze circuits comprised of those devices. The course will also introduce the students with basic working principles of transformer, motor and generator.															

20	Course Synopsis	- DC Circuit: Electrical units and standards, Ohms law, KCL, KVL, voltage and current divider law, Series-parallel networks, Star-delta and delta-star conversion, nodal analysis, mesh analysis, Superposition theorem, Source Transformation, Thevenin's and Norton's theorem, Maximum power transfer theorem, DC analysis of energy storing element i.e. Inductor and Capacitor - AC circuit: Sinusoidal steady state analysis, Series, Parallel, and Wye-Delta AC circuits, AC circuit solving techniques (Nodal, Mesh, Superposition) and Networks theorems of AC circuits (Superposition, Thevenin's and Norton's theorem) - Electrical Machines: Basic principle of Transformer, Motor and Generator																															
21	Course Outcomes (COs)	Upon completion of this course, students will be able to: CO1: Understand basic electrical parameters such as voltage, current, power, power factor etc. and their units. CO2: Apply fundamental network theorems such as Thevenin's theorem, maximum power transfer theorem, KVL, KCL etc. to solve for different electrical quantities in both AC and DC circuits. CO3: Explain the applications of basic electrical circuit components such as resistors, capacitors, inductors etc. and the working principle of electrical machines such as transformers, motors and generators.																															
<table><tr><th colspan="7">CO Attainment Process</th></tr><tr><th>CO No.</th><th>PO No.</th><th>Bloom's Domain / Level</th><th>Knowledge Profile, K</th><th>Complex Engineering Problems, WP</th><th>Delivery Methods / Activities</th><th>Assessment Tools</th></tr><tr><td>CO1</td><td>PO1</td><td>Cognitive / Understanding</td><td>K1, K2, K3</td><td>---</td><td rowspan="3"> - Class Lecture - Audio-Visual - Home Assignment - Class Exercise - Group Discussion </td><td rowspan="3"> - Attendance & class performance - Class Test - Assignment - Mid Exam - Final Exam </td></tr><tr><td>CO2</td><td>PO2</td><td>Cognitive / Analyzing</td><td>K1, K2, K3</td><td>WP1</td></tr><tr><td>CO3</td><td>PO2</td><td>Cognitive / Analyzing</td><td>K1, K2, K3</td><td>WP1</td></tr></table>			CO Attainment Process							CO No.	PO No.	Bloom's Domain / Level	Knowledge Profile, K	Complex Engineering Problems, WP	Delivery Methods / Activities	Assessment Tools	CO1	PO1	Cognitive / Understanding	K1, K2, K3	---	- Class Lecture - Audio-Visual - Home Assignment - Class Exercise - Group Discussion	- Attendance & class performance - Class Test - Assignment - Mid Exam - Final Exam	CO2	PO2	Cognitive / Analyzing	K1, K2, K3	WP1	CO3	PO2	Cognitive / Analyzing	K1, K2, K3	WP1
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22	Teaching Strategy	Maximum topics will be covered from the textbook. For the rest of the topics, reference books will be followed. Some class notes will be uploaded on the web. White board will be used for most of the time. For some cases, multimedia projector will be used for the convenience of the students. Students must participate in classroom discussions for case studies, problems solving and project developments.																															
23	Lecture Plan	<table><tr><th>Lecture #</th><th>Selected Topics</th><th>Article (Text book)</th><th>Prob./Activities (Text book)</th><th>Course Outcome</th></tr><tr><td>1-3 (Week-1)</td><td>Definition of electrical quantities such as charge, voltage, current, power, energy etc. and their units, Introduction to electrical circuits, Active and passive electrical elements, Passive sign convention</td><td>Chapter 1</td><td>Examples: 1.1,1.2, 1.3, 1.4, 1.5, 1.6, 1.7 Practice problems: 1.4, 1.5, 1.6, 1.7</td><td>CO1</td></tr><tr><td>3-6 (Week 2-3)</td><td>Series and parallel combination of resistors, finding equivalent resistance, definition of nodes, branches and loops. KCL, KVL and current and voltage divider laws, Wye-Delta transformation</td><td>Chapter 2</td><td>Examples: 2.1, 2.2, 2.3, 2.4, 2.5, 2.6, 2.7, 2.8, 2.9, 2.10 Practice Problems: 2.1, 2.2, 2.3, 2.4 Class Test-01</td><td>CO2</td></tr><tr><td>7-11 (Week 4-6)</td><td>Nodal analysis, Mesh analysis</td><td>Chapter 3</td><td>Assignment Examples: 3.1, 3.2,3.3, 3.4</td><td>CO2</td></tr></table>	Lecture #	Selected Topics	Article (Text book)	Prob./Activities (Text book)	Course Outcome	1-3 (Week-1)	Definition of electrical quantities such as charge, voltage, current, power, energy etc. and their units, Introduction to electrical circuits, Active and passive electrical elements, Passive sign convention	Chapter 1	Examples: 1.1,1.2, 1.3, 1.4, 1.5, 1.6, 1.7 Practice problems: 1.4, 1.5, 1.6, 1.7	CO1	3-6 (Week 2-3)	Series and parallel combination of resistors, finding equivalent resistance, definition of nodes, branches and loops. KCL, KVL and current and voltage divider laws, Wye-Delta transformation	Chapter 2	Examples: 2.1, 2.2, 2.3, 2.4, 2.5, 2.6, 2.7, 2.8, 2.9, 2.10 Practice Problems: 2.1, 2.2, 2.3, 2.4 Class Test-01	CO2	7-11 (Week 4-6)	Nodal analysis, Mesh analysis	Chapter 3	Assignment Examples: 3.1, 3.2,3.3, 3.4	CO2											
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			Examples: 3.5,3.7, 3.8	
12-14 (Week 6-7)	Superposition theorem, Source transformation	Chapter 4	Examples:4.1, 4.2, 4.3,4.4, 4.5, 4.6, 4.7 Practice Problem: 4.4, 4.5, 4.6, 4.7 Class Test 02	CO2
	Midterm Examination			
15-17 (Week 8-9)	Thevenin's theorem, Norton's Theorem, Maximum power transfer theorem	Chapter 4	Class Test 3 Examples: 4.8, 4.10, 4.11, 4.13 Practice problem: 4.8, 4.10, 4.11, 4.13	CO2
18-20 (Week 9-10)	6.2 Capacitors, 7.5 Step Response of an RC Circuit	Chapter 6,7	Example: 6.5, 7.10, 7.11 Practice Problems: 6.5, 7.10, 7.11	CO2, CO3
21-23 (Week11-12)	6.4 Inductors, 7.6 Step Response of an RL Circuit	Chapter 6,7	Examples: 6.10,7.12, 7.13 Practice Problems: 6.10, 7.12, 7.13	CO2, CO3
24-25 (Week 12-13)	9.2 Sinusoids, 9.3 Phasors, 9.5 Impedance and Admittance	Chapter 9	Examples: 9.1, 9.2, 9.3, 9.9 Practice Problems: 9.1, 9.2, 9.3, 9.9	CO2
25-29 (Week 13-15)	10.2 Nodal Analysis, 10.3 Mesh Analysis, 10.4 Superposition Theorem, 10.5 Source Transformation, 10.6 Thevenin and Norton Equivalent Circuit	Chapter 10	Examples: 10.1, 10.2, 10.3, 10.5, 10.6, 10.7, 10.8, 10.9 Practice Problem: 10.1, 10.2, 10.3, 10.5, 10.6, 10.7, 10.8, 10.9	CO2
30 (Week 15)	11.2 Instantaneous and Average Power 11.3 Maximum Average Power Transfer	Chapter 11	Class Test 4 Examples: 11.1, 11.2, 11.3, 11.5, 11.6 Practice Problems: 11.1, 11.2, 11.3, 11.5, 11.6	CO2
31-32 (Week 16)	Faraday's first and second laws, working principle of motor, generator; Magnetically coupled circuits	Chapter 13	Slides	CO3
	Final Exam			
24.	Assessment and Marks Distribution:	Students will be assessed on the basis of their overall performance in all the exams, quizzes, and class participation. Final evaluation will be the compilation of: ❖ Class Tests (15%) ❖ Assignment and Presentation (10%)		

		❖ Attendance and Performance (5%) ❖ Mid-Term Test (30%) ❖ Final Exam (40%)																																											
25.	Overall CO Assessment Criteria	Assessment methods of COs are given below:																																											
		<table><tr><th rowspan="2">Assessment Area</th><th colspan="3">CO</th><th rowspan="2">Assessment Area Mark</th></tr><tr><th>CO1</th><th>CO2</th><th>CO3</th></tr><tr><td>Attendance and Performance</td><td>-</td><td>-</td><td>-</td><td>-</td></tr><tr><td>Assignment and Presentation</td><td>-</td><td>10</td><td>10</td><td>20</td></tr><tr><td>Class Test</td><td>15</td><td>15</td><td>15</td><td>45</td></tr><tr><td>Midterm Exam</td><td>15</td><td>15</td><td>-</td><td>30</td></tr><tr><td>Final Exam</td><td>-</td><td>20</td><td>20</td><td>40</td></tr><tr><td>Total Mark</td><td>30</td><td>60</td><td>45</td><td>135</td></tr></table>						Assessment Area	CO			Assessment Area Mark	CO1	CO2	CO3	Attendance and Performance	-	-	-	-	Assignment and Presentation	-	10	10	20	Class Test	15	15	15	45	Midterm Exam	15	15	-	30	Final Exam	-	20	20	40	Total Mark	30	60	45	135
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26.	Rubrics	<table><tr><th>COs (Bloom's Level)</th><th>Excellent (80%-100%)</th><th>Good (70%-79%)</th><th>Satisfactory (60%-69%)</th><th>Poor (40%-59%)</th><th>Excellent (80%-100%)</th><th>Unsatisfactory (0%-39%)</th></tr><tr><td>CO1 (Understanding)</td><td>Answer is complete and sufficient detail provided to support issues related to the question. And also deals fully with the entire question.</td><td>Answer is brief with sufficient detail provided to support issues were introduced. And most of the basic details are included but some are missing.</td><td>Answer is brief with insufficient detail provided to support issues were introduced.</td><td>Answer is incomplete and excessive discussion of unrelated issues. And serious gaps in the basic details.</td><td>None of the relevant details were included or didn't answer.</td><td></td></tr><tr><td>CO2 (Applying)</td><td>Answer is complete and sufficient detail provided to support issues related to the question. And also deals fully with the entire question.</td><td>Answer is brief with sufficient detail provided to support issues were introduced. And most of the basic details are included but some are missing.</td><td>Answer is brief with insufficient detail provided to support issues were introduced.</td><td>Answer is incomplete and excessive discussion of unrelated issues. And serious gaps in the basic details.</td><td>None of the relevant details were included or didn't answer.</td><td></td></tr><tr><td>CO3 (Designing)</td><td>The question is answered appropriately by</td><td>The question is answered briefly by applying the</td><td>The question is answered correctly by applying the</td><td>The question is answered incompletely by</td><td>No attempt to implement the</td><td></td></tr></table>						COs (Bloom's Level)	Excellent (80%-100%)	Good (70%-79%)	Satisfactory (60%-69%)	Poor (40%-59%)	Excellent (80%-100%)	Unsatisfactory (0%-39%)	CO1 (Understanding)	Answer is complete and sufficient detail provided to support issues related to the question. And also deals fully with the entire question.	Answer is brief with sufficient detail provided to support issues were introduced. And most of the basic details are included but some are missing.	Answer is brief with insufficient detail provided to support issues were introduced.	Answer is incomplete and excessive discussion of unrelated issues. And serious gaps in the basic details.	None of the relevant details were included or didn't answer.		CO2 (Applying)	Answer is complete and sufficient detail provided to support issues related to the question. And also deals fully with the entire question.	Answer is brief with sufficient detail provided to support issues were introduced. And most of the basic details are included but some are missing.	Answer is brief with insufficient detail provided to support issues were introduced.	Answer is incomplete and excessive discussion of unrelated issues. And serious gaps in the basic details.	None of the relevant details were included or didn't answer.		CO3 (Designing)	The question is answered appropriately by	The question is answered briefly by applying the	The question is answered correctly by applying the	The question is answered incompletely by	No attempt to implement the											
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			applying the suggested method in the question.	suggested method in the question.	suggested method in the question but some steps are missing.	applying the suggested method in the question but some steps are correct.	suggested method.																						
27.	Grading Policy	The following chart will be followed for grading. This has been customized from the guideline provided by the School of Engineering and Computer Science.																											
		<table><tr><td>A+</td><td>A</td><td>A-</td><td>B+</td><td>B</td><td>B-</td><td>C+</td><td>C</td><td>D</td><td>F</td></tr><tr><td>≥ 80</td><td>75-<80</td><td>70-<75</td><td>65-<70</td><td>60-<65</td><td>55-<60</td><td>50-<55</td><td>45-<50</td><td>40-<45</td><td><40</td></tr></table>								A+	A	A-	B+	B	B-	C+	C	D	F	≥ 80	75-<80	70-<75	65-<70	60-<65	55-<60	50-<55	45-<50	40-<45	<40
A+	A	A-	B+	B	B-	C+	C	D	F																				
≥ 80	75-<80	70-<75	65-<70	60-<65	55-<60	50-<55	45-<50	40-<45	<40																				
28 .	Additional Course Policies	Assignments	There will be four assignments. Average marks of the assignments will be counted. No late homework will be accepted. <i>Any kind of copy/manipulation in assignment will carry zero mark.</i> Two or more copied assignments will carry zero mark in all assignments. Zero tolerance will be shown in this regard. Solutions to assignment problems will be provided through web and on hand.																										
		Class Test	There will be at least three class tests (CT). Best two of three or best three of four CTs will be counted. Both of regular and surprise CTs can be conducted.																										
		Exams	CT, Mid-term and final exam will be closed book, closed notes. Mobile phone is strictly prohibited in exam hall. Students are insisted to carry their own watch and synchronize time during exam hours.																										
		Test Policy	If a student is absent from class test anyway and made no report to the class teacher personally beforehand, his/her score for that test will be zero. No make-up for the class test will be allowed as 2 of 3 or 3 of 4 CTs are being considered.																										
29 .	Additional Information	a. Academic Calendar Summer 2020: https://bubt.edu.bd/downloads/index/9091 b. Academic Policies: https://www.bubt.edu.bd/Home/page_details/Rules_and_Regulations c. Grading & Evaluation: https://www.bubt.edu.bd/Home/page_details/Evaluation_Grading_System d. Proctorial Rules: https://www.bubt.edu.bd/Home/page_details/Office_of_the_Proctor																											
30 .	Descriptions of Cognitive Domain (Anderson and Krathwohl's Taxonomy 2001)	<table><tr><td colspan="4">The cognitive domain involves the development of our mental skills and the acquisition of knowledge.</td></tr><tr><td>Level</td><td>Category</td><td>Meaning</td><td>Keywords</td></tr><tr><td>C1</td><td>Remembering</td><td>Recognizing or recalling knowledge from memory. Remembering is when memory is used to produce or retrieve definitions, facts, or lists, or to recite previously learned information.</td><td>Define, describe, draw, find, identify, label, list, match, name, quote, recall, recite, tell, write</td></tr><tr><td>C2</td><td>Understanding</td><td>Constructing meaning from different types of functions be they written or graphic messages or activities like interpreting, exemplifying, classifying, summarizing, inferring, comparing, or explaining.</td><td>Classify, compare, exemplify, conclude, demonstrate, discuss, explain, identify, illustrate, interpret, paraphrase, predict, report</td></tr><tr><td>C3</td><td>Applying</td><td>Carrying out or using a procedure through executing, or implementing. Applying relates to or refers to situations where learned material is used through products like models, presentations, interviews or simulations.</td><td>Apply, change, choose, compute, dramatize, implement, interview, prepare, produce, role play, select, show, transfer, use</td></tr></table>								The cognitive domain involves the development of our mental skills and the acquisition of knowledge.				Level	Category	Meaning	Keywords	C1	Remembering	Recognizing or recalling knowledge from memory. Remembering is when memory is used to produce or retrieve definitions, facts, or lists, or to recite previously learned information.	Define, describe, draw, find, identify, label, list, match, name, quote, recall, recite, tell, write	C2	Understanding	Constructing meaning from different types of functions be they written or graphic messages or activities like interpreting, exemplifying, classifying, summarizing, inferring, comparing, or explaining.	Classify, compare, exemplify, conclude, demonstrate, discuss, explain, identify, illustrate, interpret, paraphrase, predict, report	C3	Applying	Carrying out or using a procedure through executing, or implementing. Applying relates to or refers to situations where learned material is used through products like models, presentations, interviews or simulations.	Apply, change, choose, compute, dramatize, implement, interview, prepare, produce, role play, select, show, transfer, use
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			C4	Analyzing	Breaking materials or concepts into parts, determining how the parts relate to one another or how they interrelate, or how the parts relate to an overall structure or purpose. Mental actions included in this function are differentiating, organizing, and attributing, as well as being able to distinguish between the components or parts. When one is analyzing, he/she can illustrate this mental function by creating spreadsheets, surveys, charts, or diagrams, or graphic representations.	Analyze, characterize, classify, compare, contrast, debate, deconstruct, deduce, differentiate, discriminate, distinguish, examine, organize, outline, relate, research, separate, structure
			C5	Evaluating	Making judgments based on criteria and standards through checking and critiquing. Critiques, recommendations, and reports are some of the products that can be created to demonstrate the processes of evaluation.	Appraise, argue, assess, choose, conclude, critique, decide, evaluate, judge, justify, predict, prioritize, prove, rank, rate, select, Monitor
			C6	Creating	Putting elements together to form a coherent or functional whole; reorganizing elements into a new pattern or structure through generating, planning, or producing. Creating requires users to put parts together in a new way, or synthesize parts into something new and different creating a new form or product. This process is the most difficult mental function.	Construct, design, develop, generate, hypothesize, invent, plan, produce, compose, create, make, perform, plan, produce
31	•	Descriptions of Affective Domain (Krathwohl, Bloom, Masia, 1973)	The affective domain includes the manner in which we deal with things emotionally, such as feelings, values, appreciation, enthusiasm, motivations, and attitudes.			

				of specified values, while clues to these values are expressed in the learners' over behavior and are often identifiable.																					
		A4	Organizing	Organizes values into priorities by contrasting different values, resolving conflicts between them, and creating a unique value system. The emphasis is on comparing, relating, and synthesizing values.	Compares, relates, synthesizes																				
		A5	Characterizing	Has a value system that controls their behavior. The behavior is a pervasive, consistent, predictable, and most important characteristic of the learner. Instructional objectives are concerned with the student's general patterns of adjustment (Personal, Social, Emotional).	Acts, discriminates, display influences, modifies, performs, qualifies, questions, revises, serves solves, verifies																				
32	•	<table><tr><td colspan="4">The psychomotor domain includes physical movement, coordination, and use of the motor skill areas. Development of these skills requires practice and is measured in terms of speed, precision, distance, procedures, or techniques in execution.</td></tr><tr><td>Level</td><td>Category</td><td>Meaning</td><td>Keywords</td></tr><tr><td>P1</td><td>Perception</td><td>The ability to use sensory cues to guide motor activity. This ranges from sensory stimulation, through cue selection, to translation.</td><td>Choose, describes, detects differentiates, distinguishes, identifies, isolates, relates, selects.</td></tr><tr><td>P2</td><td>Set</td><td>Readiness to act. It includes mental, physical, and emotional sets. These three sets are dispositions that predetermine a person's response to different situations (sometimes called mindsets)</td><td>Begins, displays, explains, moves, proceeds, reacts, shows, states, volunteers.</td></tr><tr><td>P3</td><td>Guided Response</td><td>The early stages in</td><td>Copies, traces, follows,</td></tr></table>				The psychomotor domain includes physical movement, coordination, and use of the motor skill areas. Development of these skills requires practice and is measured in terms of speed, precision, distance, procedures, or techniques in execution.				Level	Category	Meaning	Keywords	P1	Perception	The ability to use sensory cues to guide motor activity. This ranges from sensory stimulation, through cue selection, to translation.	Choose, describes, detects differentiates, distinguishes, identifies, isolates, relates, selects.	P2	Set	Readiness to act. It includes mental, physical, and emotional sets. These three sets are dispositions that predetermine a person's response to different situations (sometimes called mindsets)	Begins, displays, explains, moves, proceeds, reacts, shows, states, volunteers.	P3	Guided Response	The early stages in	Copies, traces, follows,
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	Descriptions of Psychomotor Domain (Simpson, 1972)																								

				learning a complex skill that includes imitation and trial and error. Adequacy of performance is achieved by practicing.	react reproduces, responds.
		P4	Mechanism	This is the intermediate stage in learning a complex skill. Learned responses have become habitual and the movements can be performed with some confidence and proficiency.	Assembles, calibrates, constructs, dismantles, displays, fastens, fixes, grinds, heats, manipulates, measures, mends mixes, organizes, and sketches.
		P5	Complex overt Response	The skillful performance of motor acts that involve complex movement patterns. Proficiency is indicated by a quick, accurate, and highly coordinated performance, requiring a minimum of energy. This category includes performing without hesitation, and automatic performance.	Assembles, builds, calibrates, constructs, dismantles, displays, fastens, fixes, grinds, heats, manipulates, measures, mends, mixes, organizes, and sketches.
		P6	Adaptation	Skills are well developed and the individual can modify movement patterns to fit special requirements.	Adapts, alters, changes, rearranges, reorganizes, revises, varies.
		P7	Origination	Creating new movement patterns to fit a particular situation or specific problem. Learning outcomes emphasize creativity based upon highly developed skills.	Arranges, builds, combines, composes, constructs, creates, designs, initiate, makes, originates.
33 .	Bloom's Taxonomy for Teaching-Learning				

Bloom's Taxonomy is a set of three hierarchical models used to classify educational learning objectives into levels of complexity and specificity. The three lists cover the learning objectives in Cognitive, Affective and Psychomotor domains. The Cognitive domain list has been the primary focus of most education and is frequently used to structure curriculum learning objectives, assessments and activities. The three domains and respective levels are illustrated below.

Cognitive [C] (Knowledge-based)	Affective [A] (Emotion-based)	Psychomotor [P] (Action-based)
1. Remembering	1. Receiving	1. Imitating
2. Understanding	2. Responding	2. Manipulating
3. Applying	3. Valuing	3. Précising
4. Analyzing	4. Organizing	4. Articulating
5. Evaluating	5. Characterizing	5. Naturalizing
6. Creating	--- --- ---	--- --- ---

34 Graduate Attributes (Program Outcomes) for B.Sc. in Engineering Program based on Washington Accord

Program Outcomes (POs) are narrower statements that describe what students are expected to know and be able to do by the Time of graduation. These relate to the knowledge skills and attitudes that students acquire while progressing through the program. The students of the B.Sc. in EEE program are expected to achieve the following graduate attributes or program outcomes at the time of graduation.

PO1–Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals and an engineering specialization to the solution of complex engineering problems.

PO2–Problem analysis: Identify, formulate, research the literature and analyze complex engineering problems and reach substantiated conclusions using first principles of mathematics, the natural sciences and the engineering sciences.

PO3–Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for public health and safety as well as cultural, societal and environmental concerns.

PO4–Investigation: Conduct investigations of complex problems, considering design of experiments, analysis and interpretation of data and synthesis of information to provide valid conclusions.

PO5–Modern tool usage: Create, select and apply appropriate techniques, resources and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO6–The engineer and society: Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practice.

PO7–Environment and sustainability: Understand the impact of professional engineering solutions in societal and environmental contexts and demonstrate the knowledge of, and need for sustainable development.

PO8–Ethics: Apply ethical principles and commit to professional ethics, responsibilities and the norms of the engineering practice.

PO9–Individual work and teamwork: Function effectively as an individual and as a member or leader of diverse teams as well as in multidisciplinary settings.

PO10–Communication: Communicate effectively about complex engineering activities with the engineering community and with society at large. Be able to comprehend and write effective reports, design documentation, make effective presentations and give and receive clear instructions.

PO11–Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work as a member or a leader of a team to manage projects in multidisciplinary environments.

PO12–Life-long learning: Recognize the need for and have the preparation and ability to engage in independent, life-long learning in the broadest context of technological change.

35 Knowledge Profile (WK)

WK1: A systematic, theory-based understanding of the natural sciences applicable to the discipline.

WK2: Conceptually-based mathematics, numerical analysis, statistics and formal aspects of computer and information science to support analysis and modeling applicable to the discipline.

WK3: A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline.

	WK4: Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline. WK5: Knowledge that supports engineering design in a practice area. WK6: Knowledge of engineering practice (technology) in the practice areas in the engineering discipline. WK7: Comprehension of the role of engineering in society and identified issues in engineering practice in the discipline: ethics and the professional responsibility of an engineer to public safety; the impacts of engineering activity: economic, social, cultural, environmental and sustainability. WK8: Engagement with selected knowledge in the research literature of the discipline.		
36 .	Range of Complex Engineering Problem Solving		
	Complex Engineering Problems have characteristic WP1 and some or all of WP2 to WP7: WP1–Depth of knowledge required: Cannot be resolved without in-depth engineering knowledge at the level of one or more of K3, K4, K5, K6 or K8 which allows a fundamentals-based, first principles analytical approach WP2–Range of conflicting requirements: Involve wide-ranging or conflicting technical, engineering and other issues WP3–Depth of analysis required: Have no obvious solution and require abstract thinking, originality in analysis to formulate suitable models WP4–Familiarity of issues: Involve infrequently encountered issues WP5–Extent of applicable codes: Are outside problems encompassed by standards and codes of practice for professional engineering WP6–Extent of stakeholder involvement and conflicting requirements: Involve diverse groups of stakeholders with widely varying needs WP7–Interdependence: Are high level problems including many component parts or sub-problems		
37 .	Range of Complex Engineering Activities		
	Attribute Complex activities means (engineering) activities or projects that have some or all of the following characteristics: EA1–Range of resources: Involve the use of diverse resources (and for this purpose resources include people, money, equipment, materials, information and technologies) EA2–Level of interaction: Require resolution of significant problems arising from interactions between wide-ranging or conflicting technical, engineering or other issues EA3–Innovation: Involve creative use of engineering principles and research based knowledge in novel ways EA4–Consequences for society and the environment: Have significant consequences in a range of contexts, characterized by difficulty of prediction and mitigation EA5–Familiarity: Can extend beyond previous experiences by applying principles-based approaches		
38 .	Social & Moral Capital		
	Our promises are based on the three cardinal principles: (a) What we do believe (b) What we do practice, and (c) What we will promote However, students are advised to undertake the following commitments for moral development.		
	1. To be punctual and attentive in class 2. To maintain inclusive learning environment 3. To ensure mutual respect 4. To be cooperative in group learning. 5. To be innovative and Creative 6. To follow dress code and wearing ID card 7. To be always proactive	8. Try to follow and review day to day class 9. To avoid conspiracy 10. To prioritize honesty & faith 11. To be motivated for asking question and encourage feedback 12. To develop attitude for speaking in English 13. Do not ignore to carry out any assignments or commitments 14. To be clean and decent in all levels.	15. To be sincere for class preparation 16. Do not forget to switch-off the cellphone in class 17. Do not forget to carry course pack and learning stuffs in class 18. To maintain loyalty and trust to the university 19. Must avoid unfair means and plagiarism in exam, reports and assignments 20. Must maintain eco-friendly environment in the campus.